

Bounded-Degree Minimum-Radius Spanning Trees for Fast Data Collection in Sensor Networks

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Abstract—We study the effect of routing topologies on the aggregated data collection rate and packet delays in TDMA-based sensor networks. We propose a bi-criteria formulation to the optimal routing tree construction problem, and show that a tree with bounded node degree and minimum radius can achieve the best trade-off in maximizing the data collection rate and minimizing the maximum packet delay.

I. INTRODUCTION

In sensor networks that require *fast* and *timely* collection of aggregated data to a common sink over a tree-based routing topology, we define *aggregated sink throughput* as the inverse of the total number of time slots required to schedule all the tree edges once per TDMA frame, and *delay* as the number of hops a packet traverses to reach the sink.

Two primary factors that directly affect the sink throughput and packet delays are: (i) efficiency of the link scheduling protocol, and (ii) structure of the routing tree. Typical sensor nodes can either transmit or receive a single packet at any given time slot due to their single half-duplex transceivers. Moreover, nodes that are very close to each other cannot be scheduled simultaneously because of wireless interference. Thus, in routing trees with high node degrees, children of a common parent need to be scheduled at different time slots, whereas trees with low node degrees might allow for more concurrent transmissions. For a given deployment, spanning trees with high node degrees have a smaller radius, and those with low node degrees have a larger radius. Therefore, packets on a low radius tree travel fewer hops to reach the sink and are likely to incur shorter delays compared to those on a tree with larger radius. Fig. 1(a) shows a shortest path tree (SPT) rooted at the sink on a network of 500 nodes randomly deployed in a region of size 200×200 . We observe that the tree has high node degrees but minimum hop distances to the sink. Fig. 1(b) shows a minimum spanning tree (MST) on the same deployment where the weight of an edge (u, v) is equal to the number of nodes covered by the union of the two disks centered at nodes u and v , each of radius equal to the Euclidean distance $d(u, v)$. This particular type of MST with edge weights defined as above is known as the minimum interference tree (MIT). Clearly, it has low node degrees but large hop distances to the sink.

The example illustrates the role of routing trees in terms of two opposing properties - *node degree* and *hop distance*

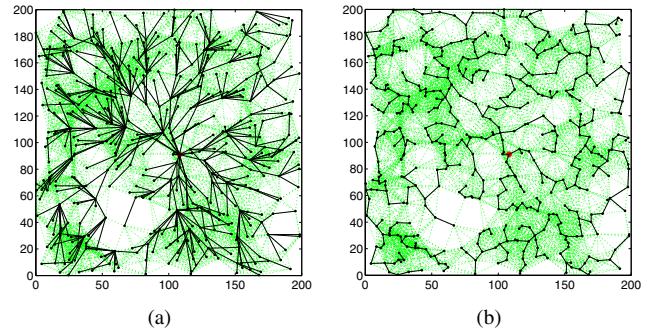


Fig. 1: (a) Shortest path tree. (b) Minimum interference tree. The sink is at the center. Dark lines represent tree edges, others represent interfering links.

- in throughput-delay performance. Given the numerous possibilities on the number of spanning trees for a large-scale network, our goal is to construct an optimal spanning tree that guarantees a bounded sink throughput as well as minimize the maximum packet delay.

II. PROBLEM FORMULATION

We model the sensor network as a graph $G = (V, E)$, where V is the set of nodes and E is the set of edges representing communication links. An edge $(u, v) \in E$ exists if $d(u, v)$ is less than the communication radius R . We denote a spanning tree on G rooted at sink $s \in V$ by $T = (V, E_T)$, where $E_T \subseteq E$ is the set of tree edges. We formulate the optimal spanning tree construction as a *bi-criteria optimization problem* [1] where, given a threshold on the maximum node degree, our goal is to minimize the maximum number of hops (radius) in the tree. We call this the *Bounded-Degree Minimum-Radius Spanning Tree* (BDMRST) problem.

DEFINITION 1. Given a constant $\Delta^* \geq 2$, the *Bounded-Degree Minimum-Radius Spanning Tree* problem is to construct a spanning tree T on G rooted at sink s , such that the radius of T is minimized subject to the constraint that the degree of any node in T is at most Δ^* .

Bi-criteria optimization problems on spanning trees are often NP-hard on general graphs, and sometimes even on geometric graphs. A slight variation of the BDMRST problem, known as the *Bounded-Degree Minimum-Diameter Spanning Tree* problem is studied in [2] for complete graphs. Extending the notion of approximation algorithms for uni-criterion

optimization problems, here our goal is to design an (α, β) -approximation to the BDMRST problem on Unit Disk Graphs, such that the maximum node degree is at most $\alpha + \Delta^*$, and the radius is at most β times the minimum possible radius subject to the degree constraint, where α, β are positive constants.

III. OPTIMAL SPANNING TREE CONSTRUCTION

Tessellate the 2-D deployment region into a set of hexagonal cells each of side length $R/2$, and associate each node to a unique cell whose center is closest to the node, breaking ties arbitrarily. Define two cells as *neighbors* if they share at least one common side, and a cell to be *adjacent* to a node u if a circle of radius R centered at u intersects the cell.

The algorithm runs in two phases, as described formally in Algorithm 1. In Phase 1, we construct a *backbone tree* $T_B = (V_B \subseteq V, E_B \subseteq E)$ by choosing one representative node r_j (called the *local root*) arbitrarily from each non-empty cell c_j , and connecting them (sometimes via *helper nodes* if a direct link does not exist) in Breadth-First-Search (BFS) order starting from the sink, making sure that the radius of T_B is not too long compared to an SPT on G . In Phase 2, we construct a *local spanning tree* within each cell from the remaining nodes in $V \setminus V_B$ lying in that cell while respecting the degree constraint Δ^* . This is always possible because the nodes within each cell form a *complete graph*. Finally, we take the union of the backbone tree and all the local spanning trees. We *mark* a cell if its local root has been included in T_B ; otherwise the cell is unmarked.

Algorithm 1 Approximation Algorithm for BDMRST

1. **Phase 1: Backbone Tree**
 2. All cells are unmarked.
 3. Initialize $T_B: V_B \leftarrow \{s\}, E_B \leftarrow \phi$, mark cell of s .
 4. Choose one local root arbitrarily from each non-empty cell; let $\mathcal{R} = \{r_1, \dots, r_n\}$ be the set of local roots.
 5. $\mathcal{Q} \leftarrow \phi$; ENQUEUE($\mathcal{Q}, s$);
 6. **while** $\mathcal{Q} \neq \phi$ **do**
 7. $u \leftarrow$ DEQUEUE($\mathcal{Q}$);
 8. **for all** unmarked cells c_j *adjacent* to u **do**
 9. $r_j \leftarrow$ local root in c_j ;
 10. **if** $d(u, r_j) \leq R$ **then**
 11. $V_B \leftarrow V_B \cup \{r_j\}; E_B \leftarrow E_B \cup \{(u, r_j)\}$;
 12. Mark c_j ; ENQUEUE($\mathcal{Q}, r_j$);
 13. **else if** $d(u, r_j) > R$ and $\exists$ helper node w **then**
 14. $V_B \leftarrow V_B \cup \{r_j, w\}$;
 15. $E_B \leftarrow E_B \cup \{(u, w), (w, r_j)\}$;
 16. Mark c_j ; ENQUEUE($\mathcal{Q}, r_j$);
 17. **Phase 2: Local Spanning Tree**
 18. **for all** non-empty cells c_j **do**
 19. $r_j \leftarrow$ local root in c_j ;
 20. Let $V_j = \{v_1 \dots v_{n_j}\}$ be the set of not yet connected nodes in c_j (V_j induces a *complete graph*).
 21. Construct local spanning tree T_j with nodes in V_j such that no node exceeds degree Δ^* .
 22. **return** $T = T_B \cup \{T_j\}$.
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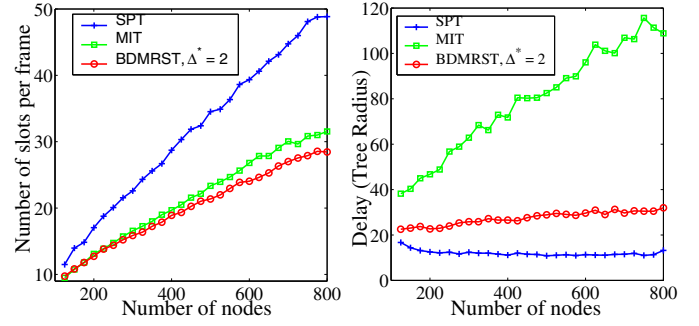


Fig. 2: Number of time slots per TDMA frame (sink throughput) and tree radius (max delay) with different network sizes for SPT, MIT, and BDMRST.

THEOREM 1. *Algorithm 1 gives a constant factor (α, β) bi-criteria approximation to the Bounded-Degree Minimum-Radius Spanning Tree problem, where $\alpha = 10$ and $\beta = 6$.*

The theorem follows from the following lemmas.

LEMMA 1. *The radius $R(T_j)$ of any local spanning tree T_j rooted at r_j and comprising $n_j + 1$ nodes $\{v_1, \dots, v_{n_j}\} \cup r_j$ is bounded by: $R(T_j) = n_j$, for $\Delta^* = 2$, and $R(T_j) \leq \log[n_j \cdot (\Delta^* - 2) + 1] + 1$, for $\Delta^* > 2$, where the logarithm is with respect to base $\Delta^* - 1$.*

LEMMA 2. *Let r_i and r_j be any two local roots on the backbone tree T_B . Let $P_G(r_i, r_j)$ be the shortest path on the original graph G between r_i and r_j consisting of m hops. Then, the length of the unique simple path $P_{T_B}(r_i, r_j)$ between r_i and r_j on T_B is at most $4m$.*

Fig. 2 plots the number of time slots required per TDMA frame to schedule all the tree edges using the interference-aware efficient link-scheduling algorithm described in [3], and the tree radius with different network sizes. We observe that a BDMRST achieves the best of both worlds in terms of high sink throughput as well as low packet delays.

IV. CONCLUSIONS

In this work, we proposed a bi-criteria approximation algorithm for constructing spanning trees that achieves a constant factor guarantee in terms of the maximum node degree and the tree radius. We showed that such a bounded-degree minimum-radius spanning tree can improve the throughput-delay trade-off for aggregated data collection in sensor networks. Our future work lies in extending this to non-uniform transmission range and scheduling with multiple frequency channels.

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